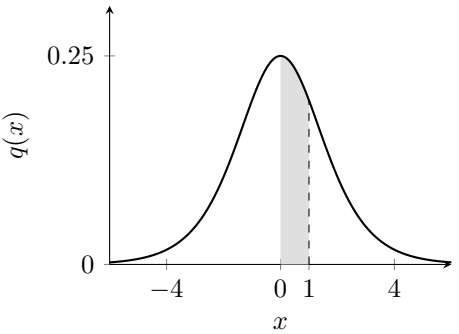
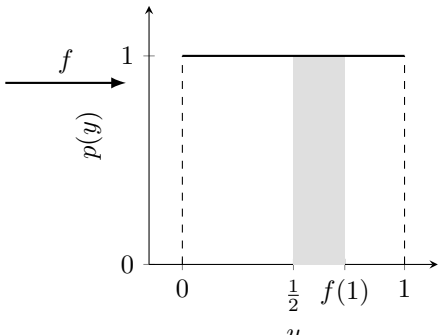
$$q(x) = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$p(y) = 1, \quad 0 < y < 1$$





$$f(x) = \frac{1}{1+e^{-x}}, \qquad f'(x) = q(x) > 0$$

$$\int_0^1 q(x)\,dx = f(1) - f(0) = \int_{1/2}^{f(1)} p(y)\,dy$$